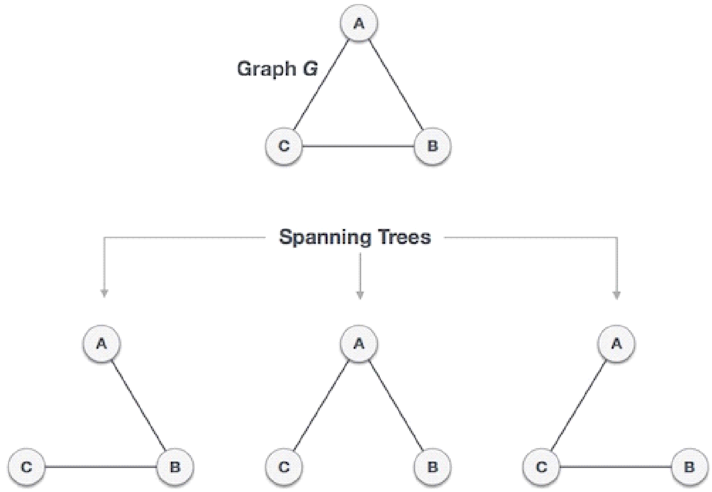
Spanning Tree

A spanning tree is a subset of Graph G, which has all the vertices covered with minimum possible number of edges. Hence, a spanning tree does not have cycles and it cannot be disconnected.

By this definition, we can draw a conclusion that every connected and undirected Graph G has at least one spanning tree. A disconnected graph does not have any spanning tree, as it cannot be spanned to all its vertices.



General Properties of Spanning Tree
We now understand that one graph can have more than one spanning tree. Following are a few properties of the spanning tree connected to graph G −

1. A connected graph G can have more than one spanning tree.
2. All possible spanning trees of graph G, have the same number of edges and vertices.
3. The spanning tree does not have any cycle (loops).
4. Removing one edge from the spanning tree will make the graph disconnected, i.e. the spanning tree is minimally connected.
5. Adding one edge to the spanning tree will create a circuit or loop, i.e. the spanning tree is maximally acyclic.

Application of Spanning Tree
Spanning tree is basically used to find a minimum path to connect all nodes in a graph. Common application of spanning trees are −

1. Civil Network Planning
2. Computer Network Routing Protocol
3. Cluster Analysis

Let us understand this through a small example. Consider, city network as a huge graph and now plans to deploy telephone lines in such a way that in minimum lines we can connect to all city nodes. This is where the spanning tree comes into picture.